

CHM 120

Molecular Modeling and Lewis Structures

Final Report

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Lesson	Molecular Modeling and Lewis Structures
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Course	CHM 120
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Test Your Knowledge

Match the number of valence electrons with the correct compound.

CH ₄	1	8
PBr ₃	2	26
OF ₂	3	20
CH ₂ O	4	12
O ₃	5	18
SiCl ₄	6	32

Move the terms to complete each statement.

A model that describes the three dimensional arrangement of electrons based on electron repulsion is called the 1 model.

Two hydrogen atoms form a 2 when they have covalent bonds.

Diamond and graphite are two 3 of carbon.

Resonance 4 show the same connection of atoms but different arrangement of the bonds.

An atom that has less than eight electrons surrounding it in a compound is said to be 5 .

The number of outermost electrons is the number of 6 .

Determine the correct molecular geometry based on the number of bonds and lone pairs around the central atom.

Bonds	Lone Pairs	Geometry
2	2	1 Bent
3	0	2 Trigonal Planar
3	1	3 Trigonal Pyramidal
4	0	4 Tetrahedral

Exploration

The localized electron model assumes that a molecule is bonded through the ____.

- ☒ sharing of valence electron pairs
- ☐ transfer of valence electron pairs

Allotropes are ____.

- ☒ different structural modifications of an element
- ☐ versions of an element having the same number of neutrons but differing numbers of protons
- ☐ molecules bonded through the sharing of valence electron pairs
- ☐ typically crystalline in structure

The number of valence electrons of an atom can be found in the row number on the periodic table.

- ☐ True
- ☒ False

Magnesium has ____ valence electrons.

- ☐ 1
- ☒ 2
- ☐ 3
- ☐ 4
- ☐ 8

An example of an atom with 3 valence electrons is ____.

- ☐ sodium
- ☐ magnesium
- ☒ aluminum
- ☐ All of the above

The octet rule is based upon the observation that atoms are most stable when surrounded by how many valence electrons?

- ☐ 6
- ☐ 4
- ☐ 12
- ☒ 8

Resonance occurs when there is only one option for the Lewis structure model.

- ☒ True
- ☐ False

What atoms always obey the octet rule?

- ☐ Hydrogen, carbon, nitrogen, oxygen
- ☐ Nitrogen, oxygen, fluorine, copper
- ☒ Oxygen, fluorine, nitrogen, carbon
- ☐ Oxygen, carbon, chlorine, helium

The goal of the VSEPR model is to arrange the position of the atoms and lone pairs surrounding the central atom in a manner that minimizes ____.

- ☒ lone pairs
- ☐ the number of electrons
- ☐ repulsion
- ☐ space and size

Molecular modeling kits are useful in determining the Lewis structure of a molecule.

- ☒ True
- ☐ False

When a molecule has more than one interior atom, ____.

- ☐ multiple Lewis structures may be drawn
- ☐ the atoms are always represented by different colored balls in the modeling kit
- ☒ more than one geometry may be used to describe the molecule
- ☐ the atoms are always connected by a double bond

Exercise 1

Data Table 1: CHO₂-

# of Valence Electrons	# of Atoms and Sets of Lone Pairs Surrounding Central Atom	Structure Geometry (Name)
18	4 atoms total; carbon atom is central atom with two oxygen atoms and one hydrogen atom surrounding it; there are 0 sets of lone pairs of electrons	trigonal planar

Photo 1: Lewis Structure of CHO_2^-

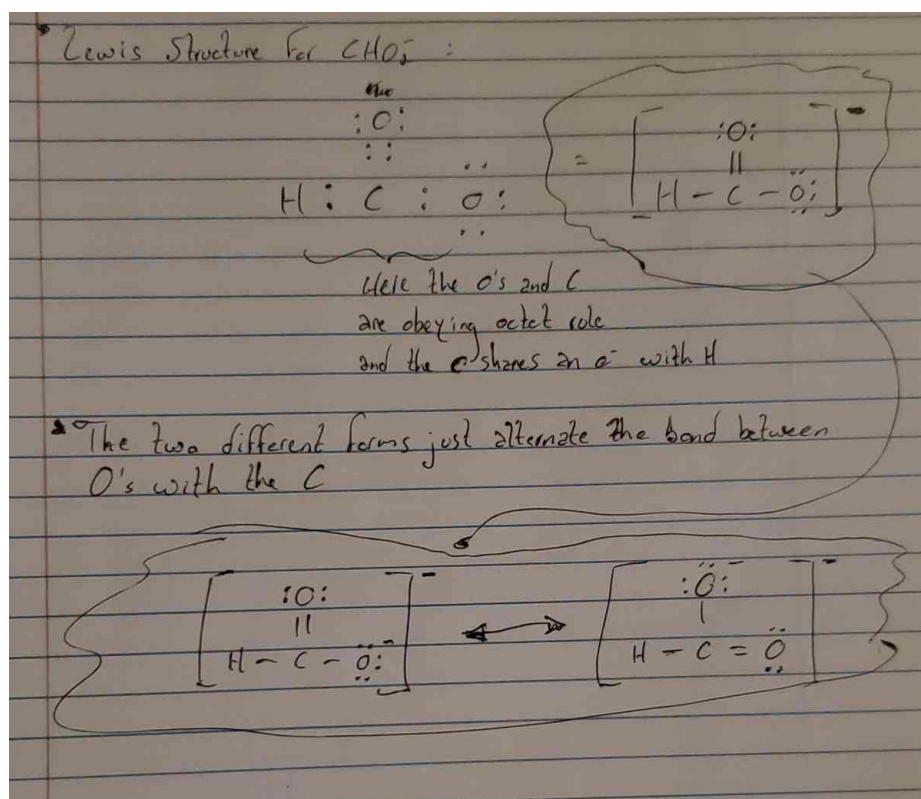
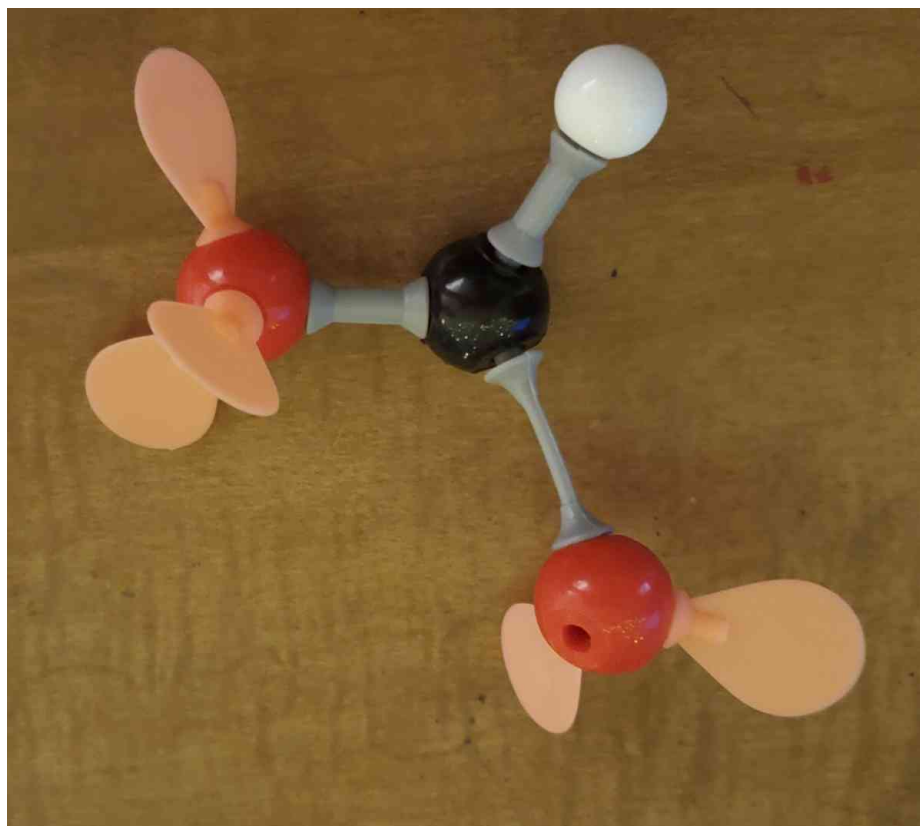


Photo 2: Molecular Model of CHO_2^-



Exercise 2

Which, if any, of the substances had resonance structures? How many resonance structures did each substance have?

B /

24 Word(s)

I believe that CHO_2^- has resonance, one resonant structure so two structures total. Also $[\text{NH}_4]^+$ and $[\text{NO}_2]^-$ have resonant structures.

$[\text{NH}_4]^+ = 4$ structures

$[\text{NO}_2]^- = 2$ structures

Both chlorine and fluorine are represented by a green modeling piece that has 4 holes. Is using the same piece for two different atoms acceptable? Why or why not?

B / U

23 Word(s)

Yes it is acceptable because they both share certain properties, especially the number of valence electrons, so the same color is okay here.

Explain why only the lone pairs on the central atom are taken into consideration when predicting molecular shape. What substances from the lab activity have more than one central atom?

B / U

66 Word(s)

All of the other electrons are occupied with their place between elements however the lone pairs are not, and perhaps provide more interest in what helps to determine the molecular shape. If the lone pairs are taken into consideration, then the chemist can start to identify in which ways the substance will grow.

The substances with more than one central atom are: H_2O_2 , C_2H_4 , CH_3OH , CH_3NH_2

Data Table 2: VSEPR Names and Atoms

Molecule or Ion	Number of Valence Electrons	Total number of atoms and sets of lone pairs surrounding central atom(s)	Structure Geometry (Name)
CCl ₄	32	5 total atoms; carbon is central atom; 0 sets of lone electron pairs	tetrahedral
H ₂ O	8	3 total atoms; oxygen is central atom; 2 sets of lone electron pairs	tetrahedral
CO ₂	8	3 total atoms; carbon is central atom; 0 sets of lone electron pairs	linear
H ₂ O ₂	14	4 total atoms; no central atom; 0 sets of lone electron pairs	tetrahedral
CH ₄	8	5 total atoms; carbon is central atom; 0 sets of lone electron pairs	tetrahedral
C ₂ H ₄	12	6 total atoms; no central atom; 0 sets of lone electron pairs	trigonal planar
CH ₂ O	12	4 total atoms; carbon is central atom; 0 sets of lone electron pairs	trigonal planar
CH ₃ OH	14	6 total atoms; carbon is central atom; 0 sets of lone electron pairs	tetrahedral
CH ₃ NH ₂	14	7 total atoms; carbon is central atom; 0 sets of lone electron pairs	tetrahedral
NF ₃	26	4 total atoms; nitrogen is central atom; 1 set of lone electron pairs	trigonal pyramidal
NH ₃	8	4 total atoms; nitrogen is central atom; 1 set of lone electron pairs	trigonal pyramidal
[NH ₄] ⁺			

	8	5 total atoms; nitrogen is central atom; 0 sets of lone electron pairs	tetrahedral
$[\text{NO}_2]^-$	18	3 total atoms; carbon is central atom; 0 sets of lone electron pairs, but one electron on central atom	bent
SO_2	18	3 total atoms; carbon is central atom; 0 sets of lone electron pairs	bent or v-shaped
OF_2	20	3 total atoms; oxygen is central atom; 2 sets of lone electron pairs	linear
H_2S	8	3 total atoms; sulfur is central atom; 2 sets of lone electron pairs	bent
SF_6	48	7 total atoms; sulfur is central atom; 0 sets of lone electron pairs	octahedral
ICl_5	42	6 total atoms; iodine is central atom; 1 set of lone electron pairs	square pyramidal

Photo 3: CCl_4 Lewis Structure and VSEPR Model

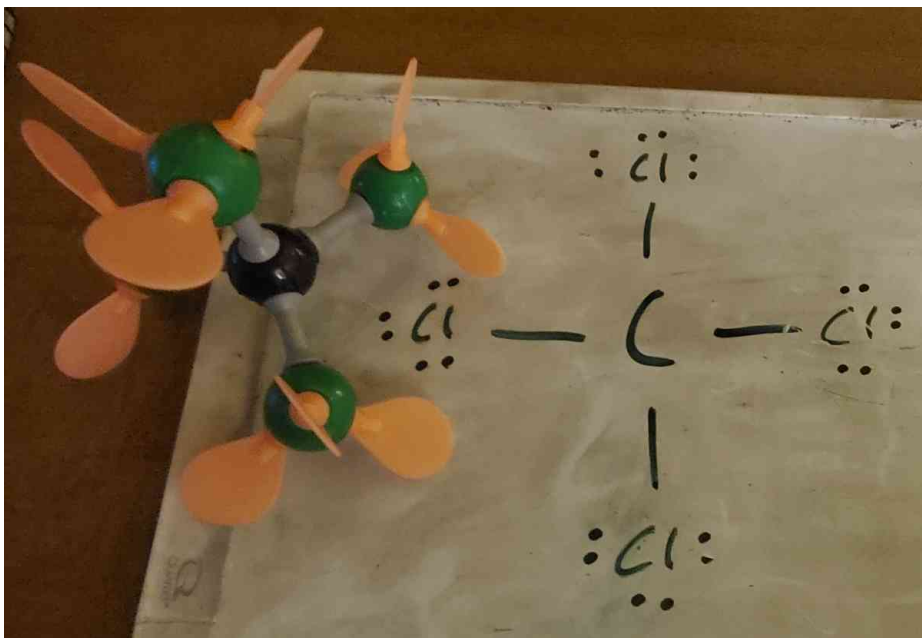


Photo 4: H₂O Lewis Structure and VSEPR Model

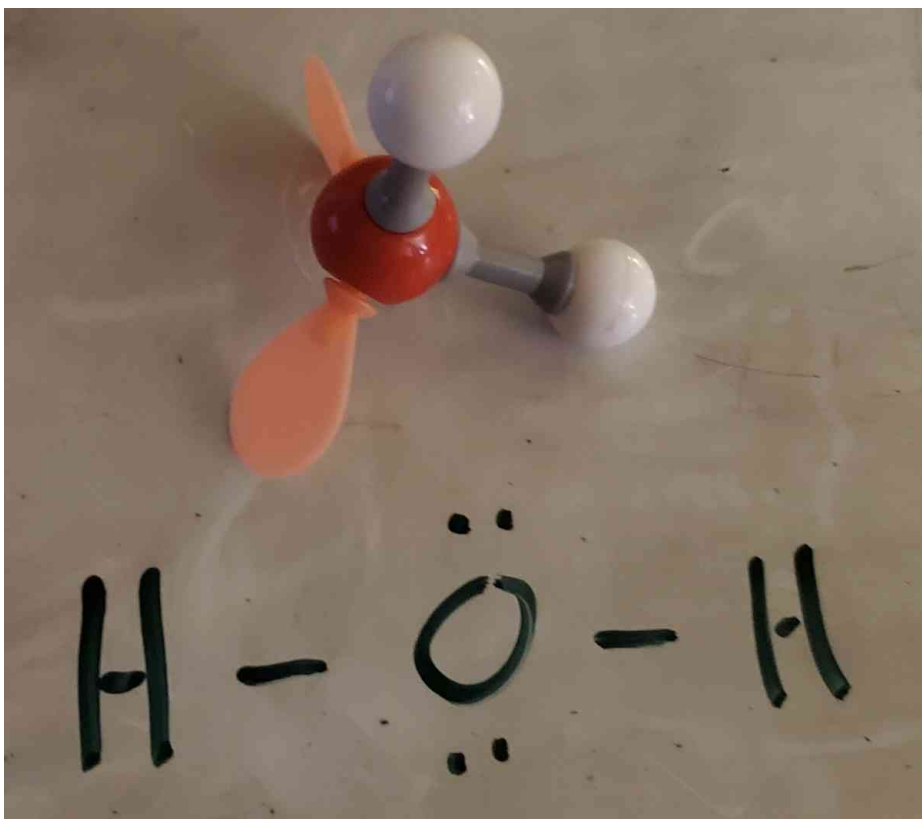


Photo 5: CO₂ Lewis Structure and VSEPR Model

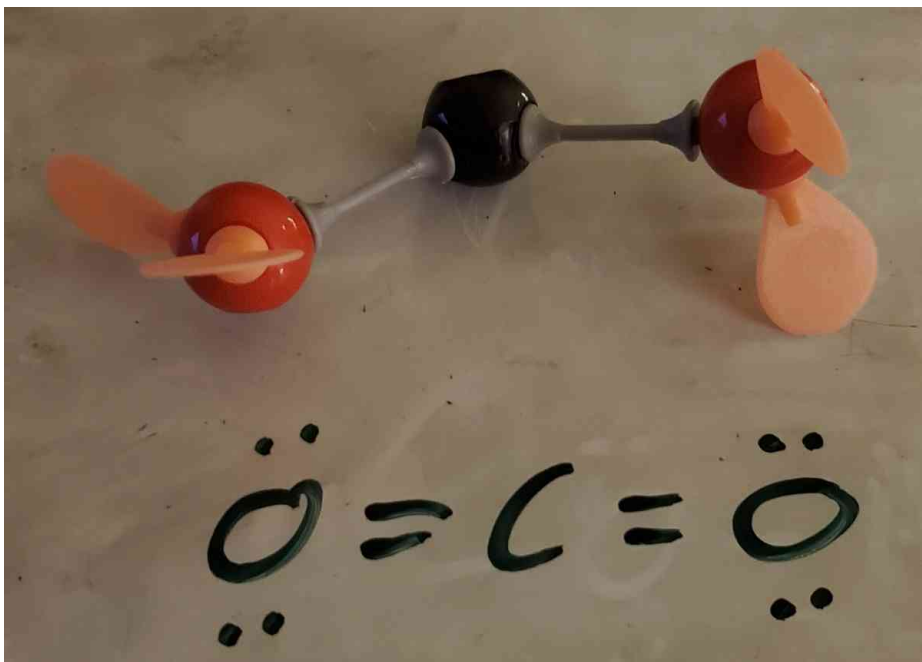


Photo 6: H₂O₂ Lewis Structure and VSEPR Model

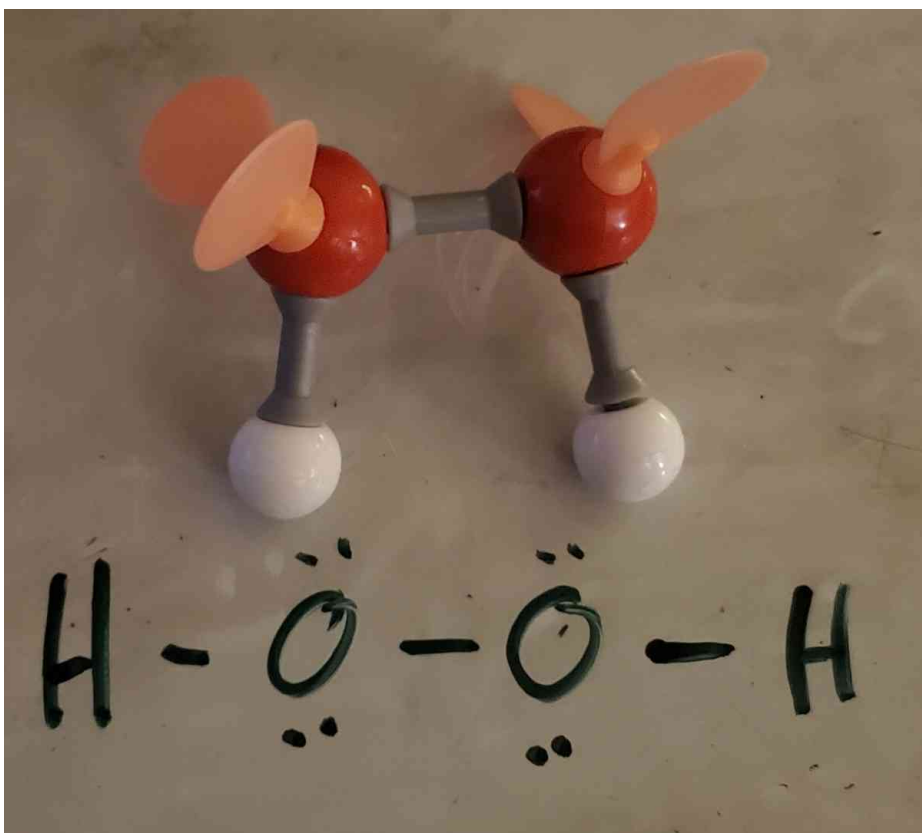


Photo 7: CH₄ Lewis Structure and VSEPR Model

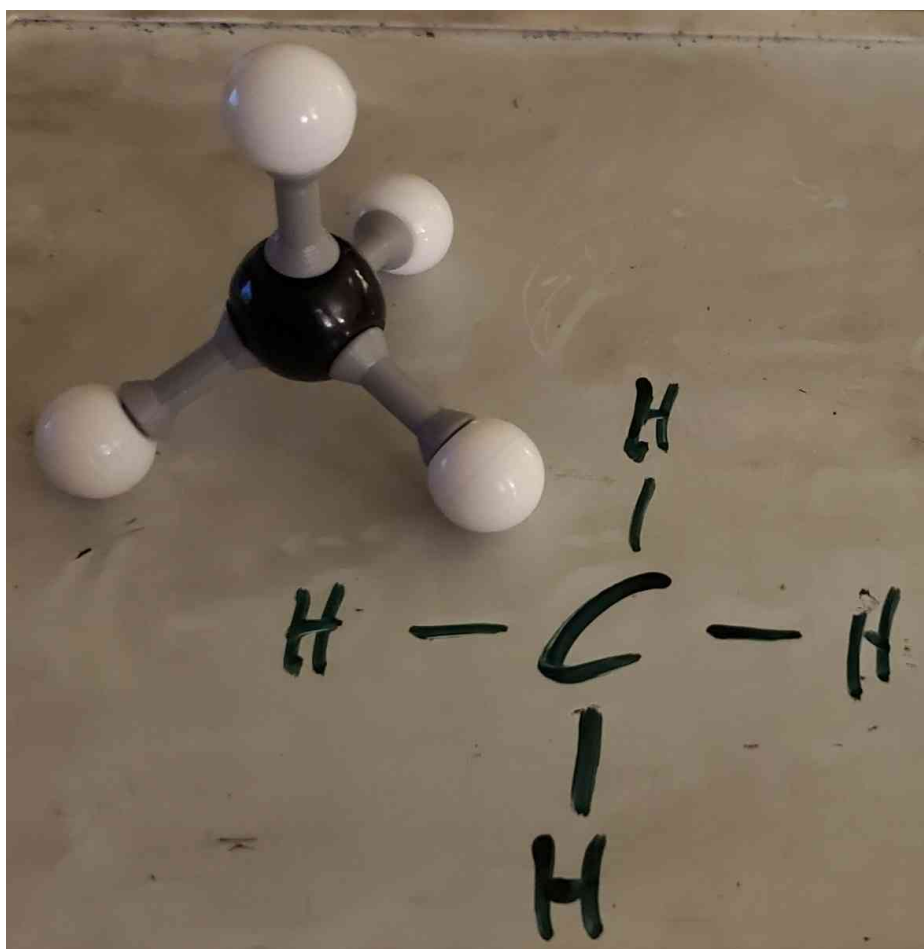


Photo 8: C₂H₄ Lewis Structure and VSEPR Model

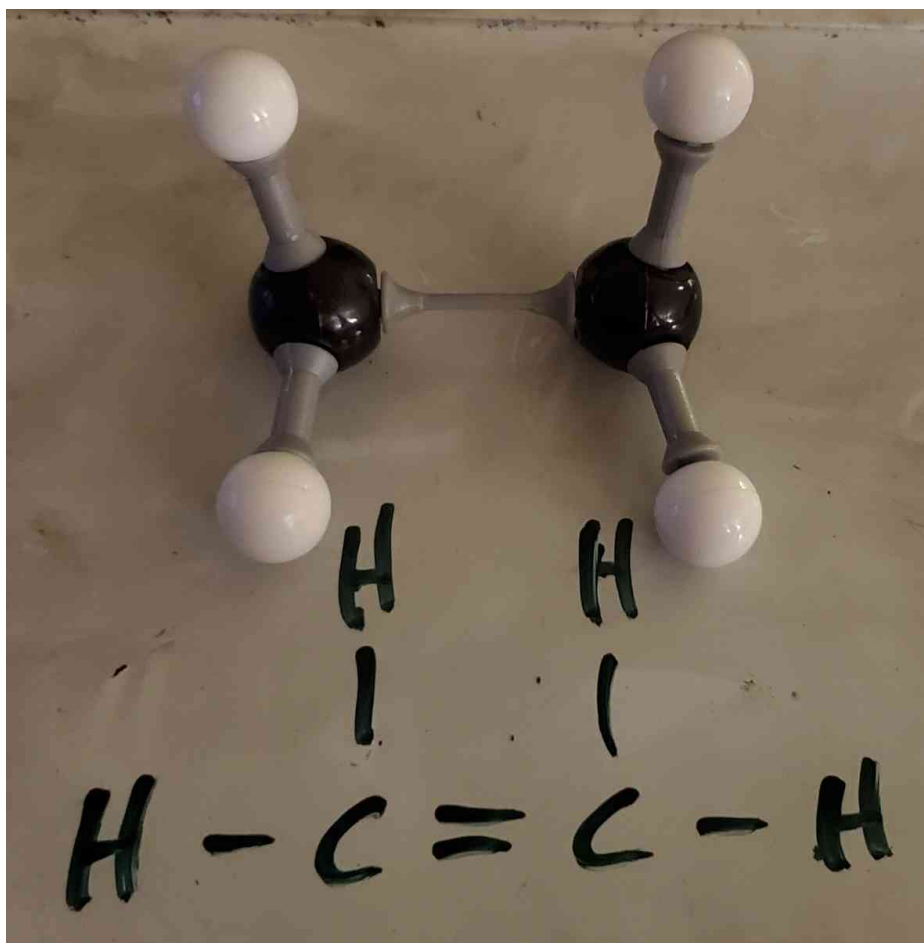


Photo 9: CH_2O Lewis Structure and VSEPR Model

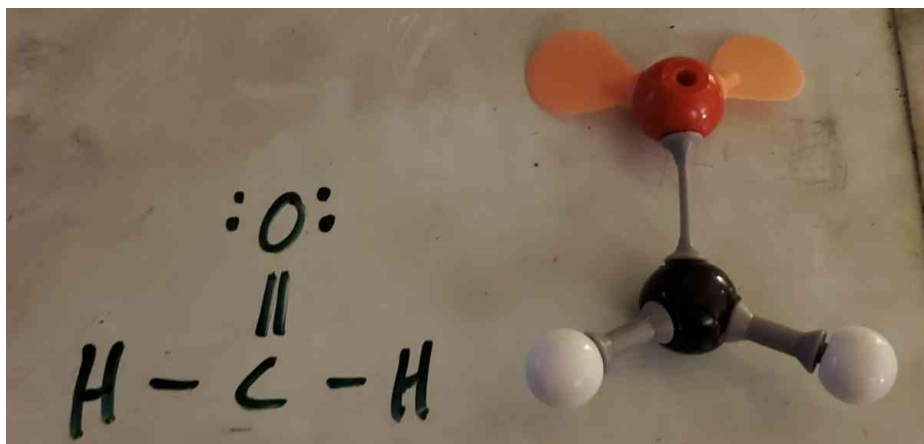


Photo 10: CH_3OH Lewis Structure and VSEPR Model

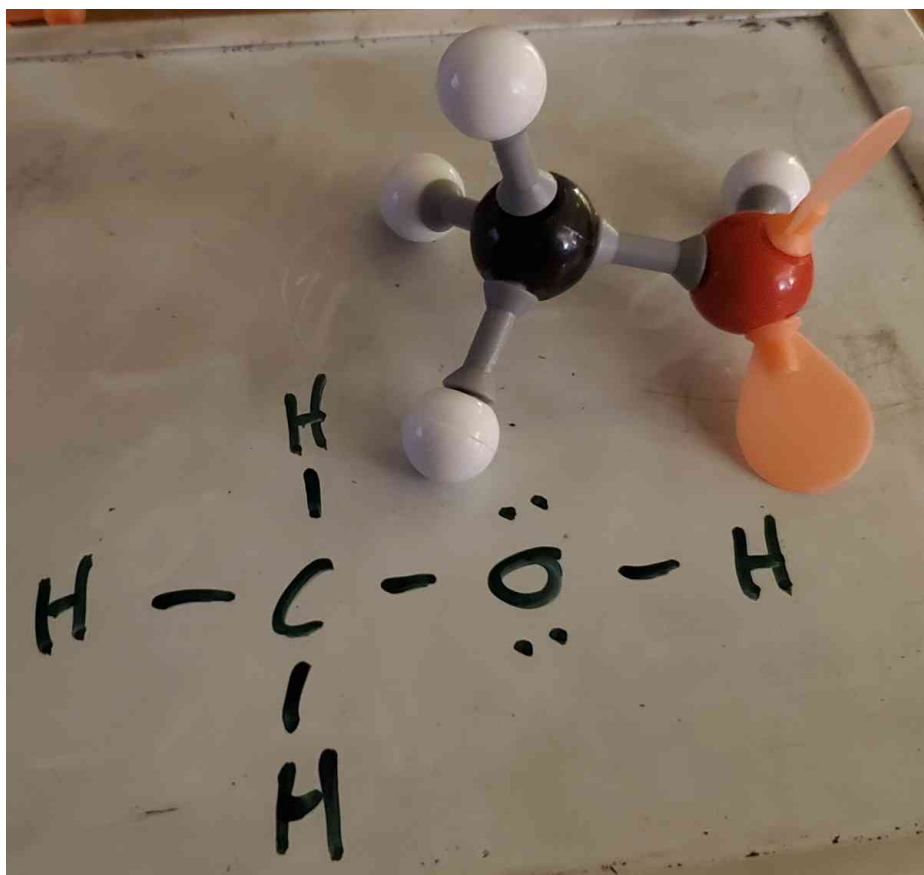


Photo 11: CH_3NH_2 Lewis Structure and VSEPR Model

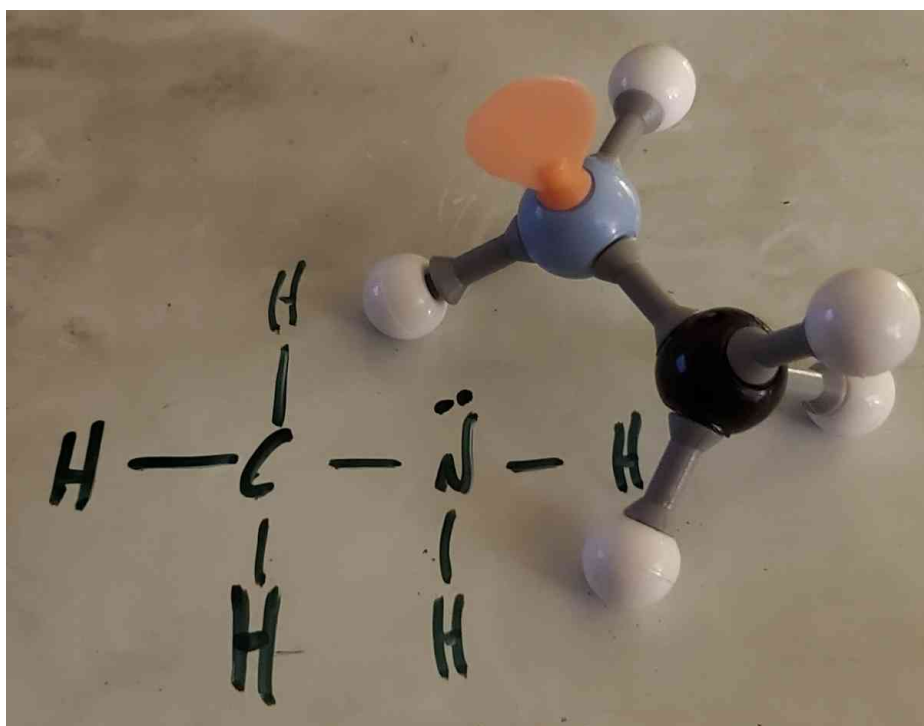


Photo 12: NF₃ Lewis Structure and VSEPR Model

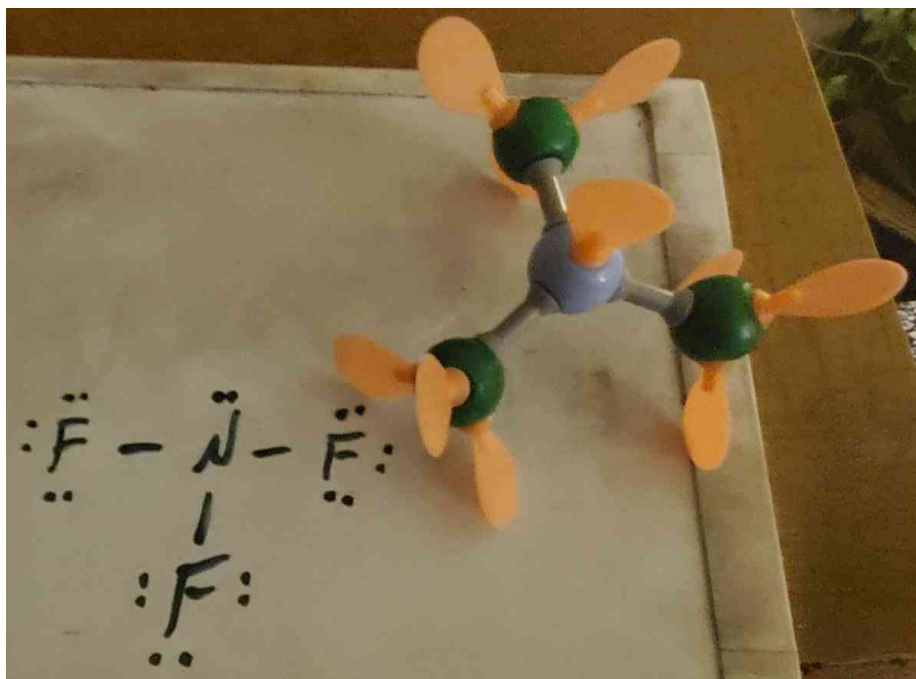


Photo 13: NH₃ Lewis Structure and VSEPR Model

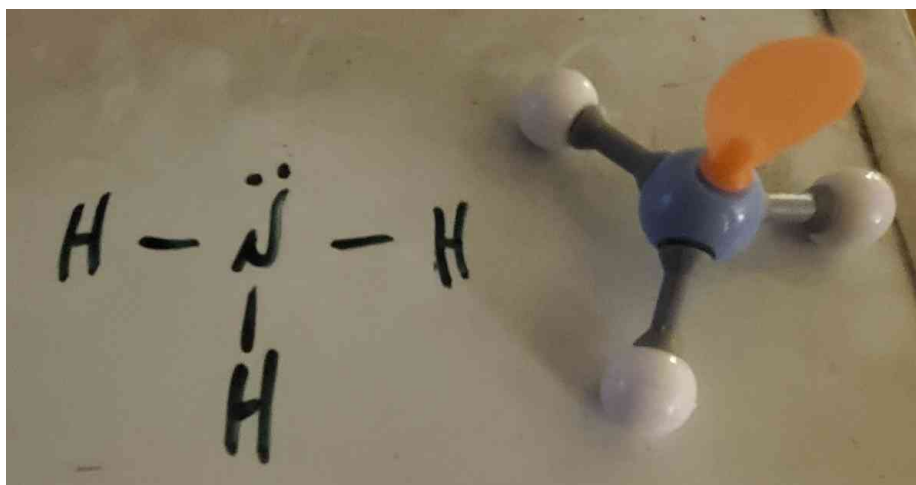


Photo 14: [NH₄]⁺ Lewis Structure and VSEPR Model

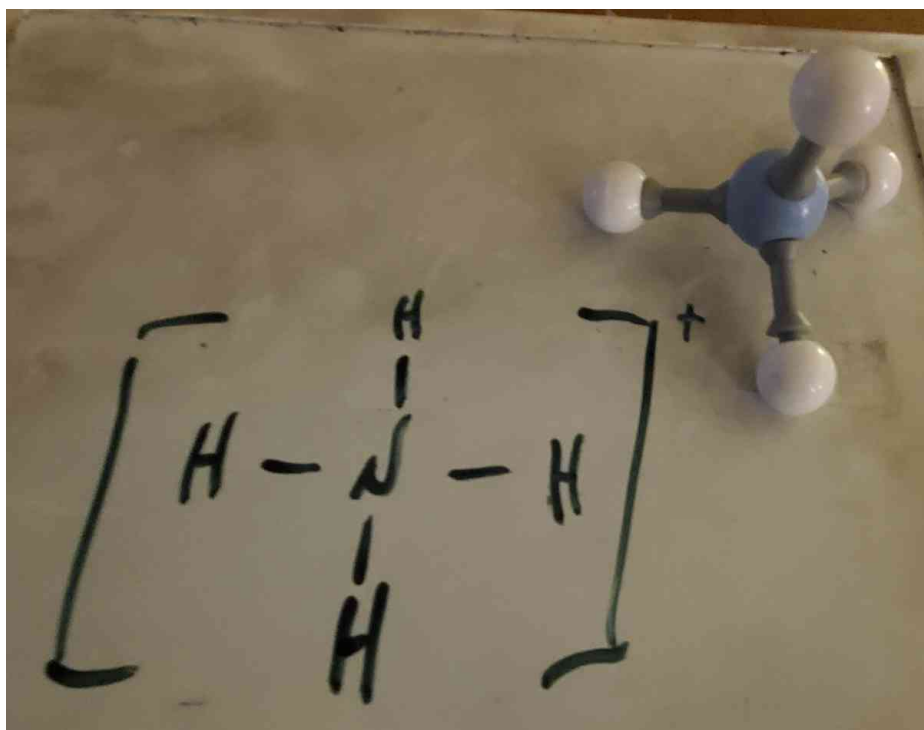


Photo 15: [NO₂]- Lewis Structure and VSEPR Model

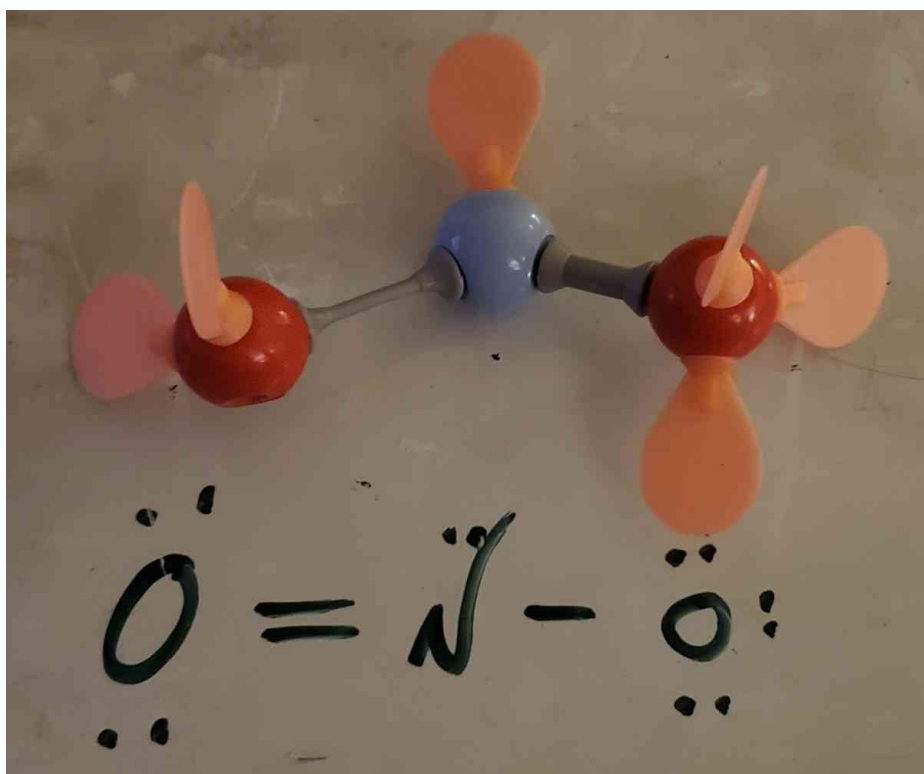


Photo 16: SO₂ Lewis Structure and VSEPR Model

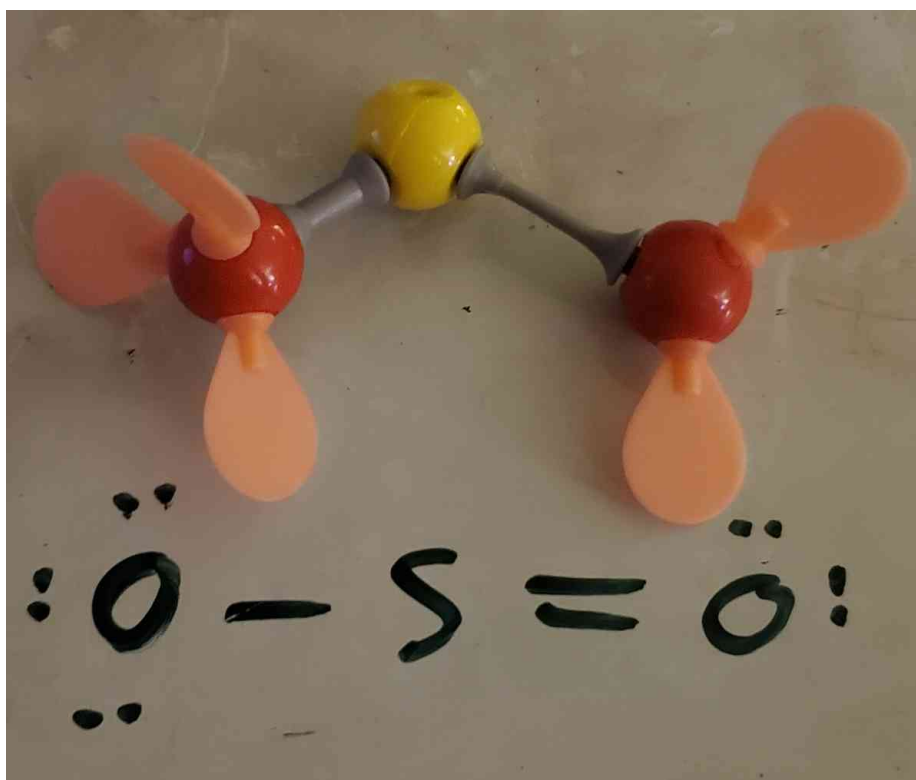


Photo 17: OF₂ Lewis Structure and VSEPR Model

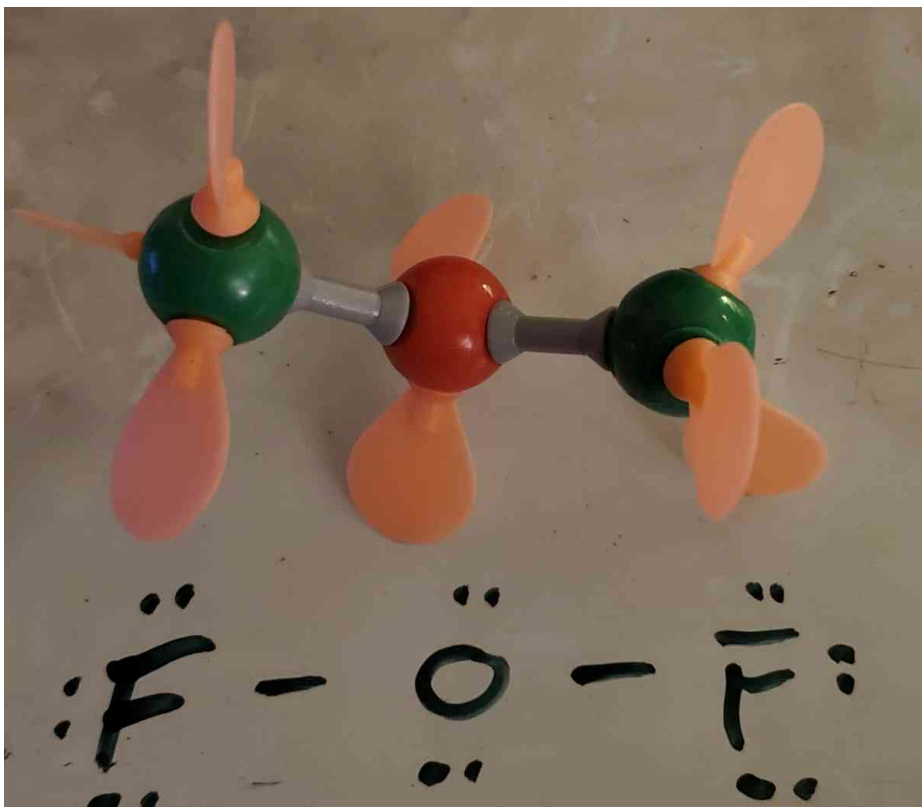


Photo 18: H₂S Lewis Structure and VSEPR Model

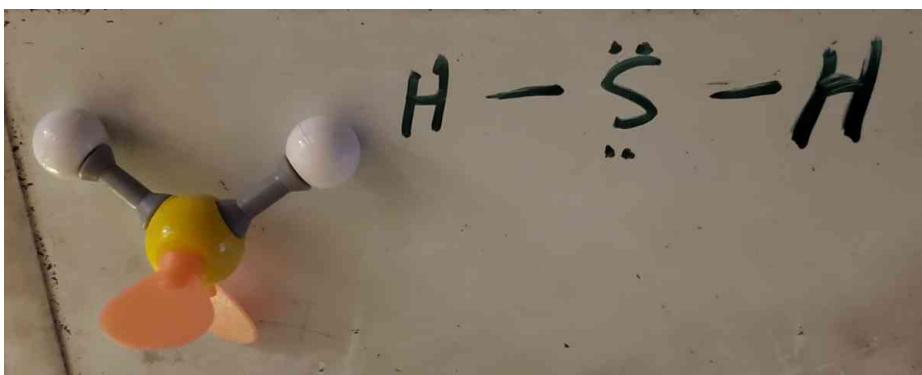


Photo 19: SF₆ Lewis Structure and VSEPR Model

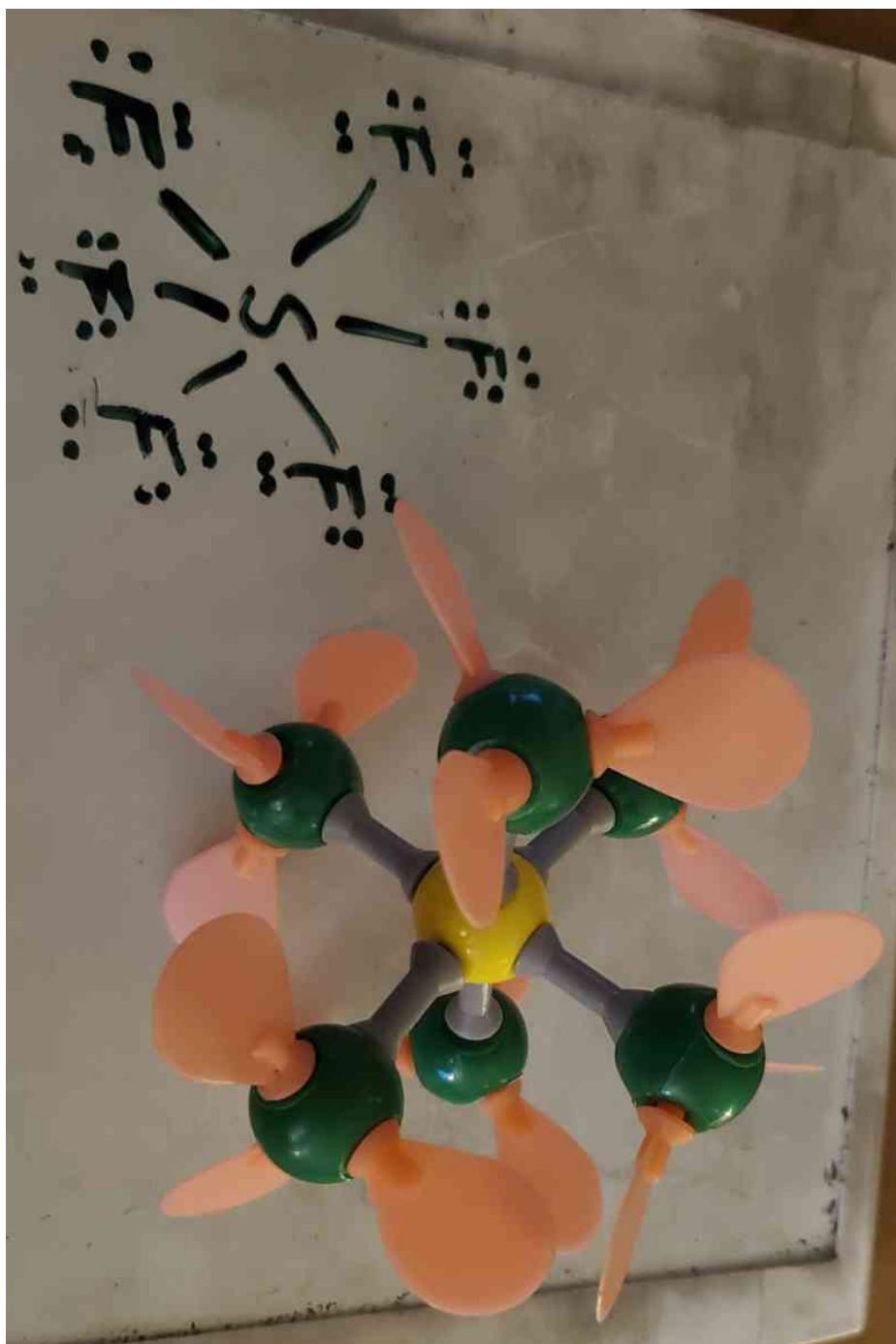
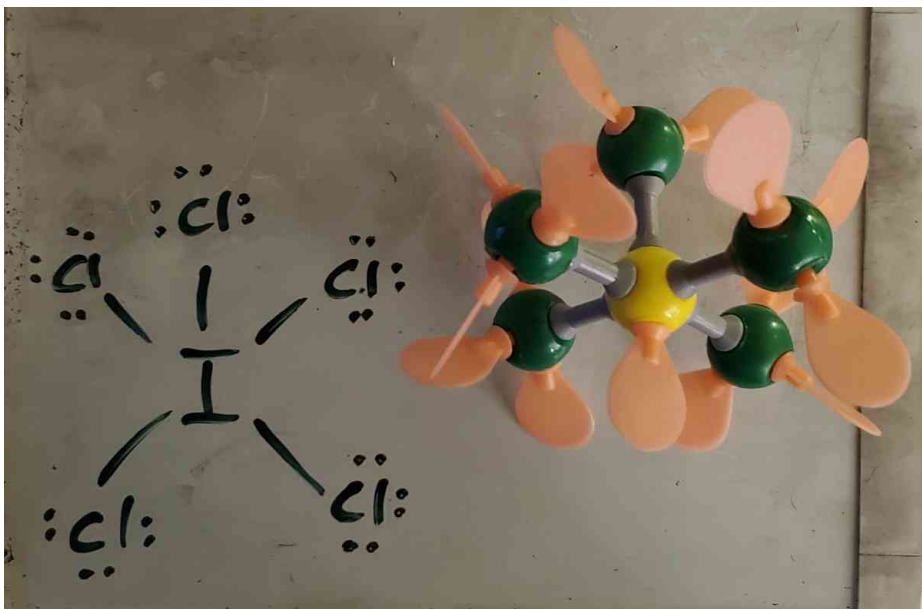


Photo 20: ICl₅ Lewis Structure and VSEPR Model



Competency Review

Different structural modifications of an element are called ____.

- ☒ allotropes
- ☐ valences
- ☐ pairs
- ☐ octets

Valence electrons include all electrons of an atom.

- ☐ True
- ☒ False

Valence electrons that are not shared between two atoms are called ____.

- ☐ doublets
- ☒ lone pairs
- ☐ octets
- ☐ electron deficient

The duet rule is based upon the observation that atoms are most stable when surrounded by eight valence electrons.

☐ True

☒ False

The ____ of a molecule visually describes the two-dimensional arrangement of atoms and valence electrons in a molecule.

☐ localized electron model

☐ charges

☒ Lewis structure

The first step in creating a Lewis structure is to determine the total number of valence electrons for the entire molecule.

☒ True

☐ False

The valence shell electron pair repulsion (VSEPR) model ____.

☒ describes the three-dimensional arrangement of the atoms in a molecule

☐ states that two negatively charged particles (electrons) will always repel one another with equal and opposite forces

☐ ensures bonding through shared valence electron pairs

☐ is the illustrated definition of the octet rule but not the duet rule

In a molecular modeling kit, a black ball with 4 holes most likely represents the element ____.

☐ hydrogen

☒ carbon

☐ magnesium

☐ sulfur

The Lewis structure for H₂O includes double bonds.

- ☐ True
- ☒ False

Some compounds have more than one Lewis structure.

- ☒ True
- ☐ False

Alkali metals are a group located at the far left of the periodic table; alkali metals have ____ valence electron(s).

- ☒ 1
- ☐ 2
- ☐ 6
- ☐ 7

Carbon dioxide has a ____ geometric structure.

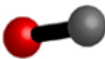
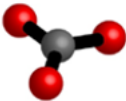
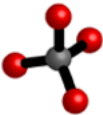
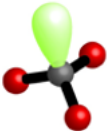
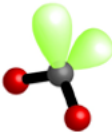
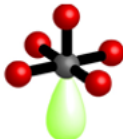
- ☐ tetrahedral
- ☐ octahedral
- ☐ trigonal planar
- ☒ linear

SF₆ has a ____ geometric structure.

- ☐ tetrahedral
- ☒ octahedral
- ☐ trigonal planar
- ☐ linear

Extension Questions

For each molecular shape, identify the number of bonds around the central atom and the number of lone pairs around the central atom.

Number of Bonds Around the Central Atom		Number of Lone Pairs Around the Central Atom		Molecular Geometries	
1		9	0		Linear
1	2		0		Linear
2	3	10	0		Trigonal planar
3	4	11	0		Tetrahedral
4	3		1		Trigonal pyramidal
5	2	12	2		Bent
6	5	13	0		Trigonal bipyramidal
7	6		0		Octahedral
8	5	14	1		Square pyramidal

